Lab Assignment: Install Parrot OS

Parrot OS is a Debian-based Linux distribution designed for security professionals, ethical hackers, and cybersecurity enthusiasts. The **Security Edition** of Parrot OS is a specialized version equipped with a wide range of pre-installed tools for penetration testing, digital forensics, reverse engineering, and vulnerability assessment. It provides a secure, lightweight, and user-friendly platform, making it an excellent choice for learning and practicing cybersecurity skills. Parrot Security is frequently updated and includes features like sandboxing and an anonymous mode for enhanced privacy.

Objectives

By the end of this lab, students will be able to:

- Download and verify the Parrot OS ISO.
- Install Parrot OS on VMware Workstation.
- Understand basic VM configuration settings.

Materials Required

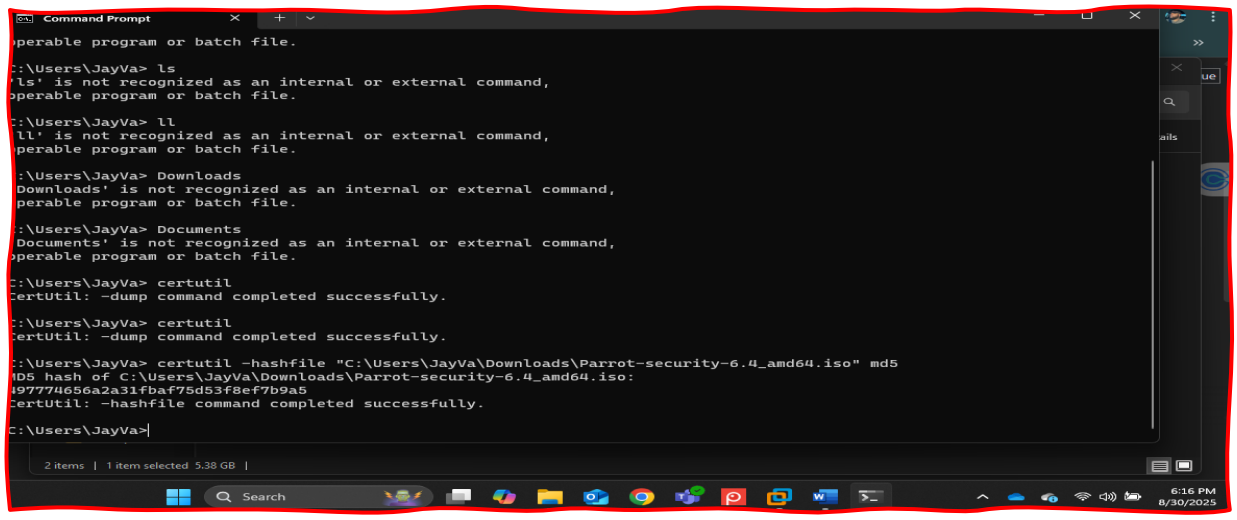
- Computer with VMware Workstation installed.
- Stable internet connection.
- At least 20 GB of free disk space.

Step 1: Download Parrot OS

1. Open a web browser and navigate to the official Parrot OS website.
2. Select the version suitable for your needs. For this lab, download **Parrot Security Edition (64-bit)**.
    - Use the following workflow to download the Parrot Security Edition (64-bit) ISO: Live → Security → AMD64 → Download
3. Save the ISO file to a location you can easily access.
4. Verify the ISO checksum (MD5/SHA256) by comparing it with the value provided on the website. You can use the following tools:
    - Windows: Use **`certutil`** in the Command Prompt.

```
Certutil -hashfile [file name] md5
```

5. Take a screenshot of the checksum:



Part 2: VMware Workstation Setup

Step 1: Create a New Virtual Machine

1. Open VMware Workstation.
2. Click on **File > New Virtual Machine**.
3. Select **Typical (recommended)** and click **Next**.
4. Choose **Installer disc image file (ISO)** and browse to the Parrot OS ISO you downloaded.
5. Select **Linux** as the guest operating system and choose **Debian 10.x 64-bit** as the version.

Step 2: Configure VM Settings

1. Assign the virtual machine a name, e.g., "Parrot OS VM."
2. Specify the disk size (minimum 20 GB) and choose **Store virtual disk as a single file**.
3. Allocate resources:
   - **Memory**: At least 2 GB (4 GB recommended).
   - **Processors**: 2 cores.
4. Ensure the network adapter is set to **NAT** for internet access.

Part 3: Installing Parrot OS

Step 1: Boot from the ISO

1. Power on the VM.
2. When prompted, choose **Install Parrot** from the boot menu.

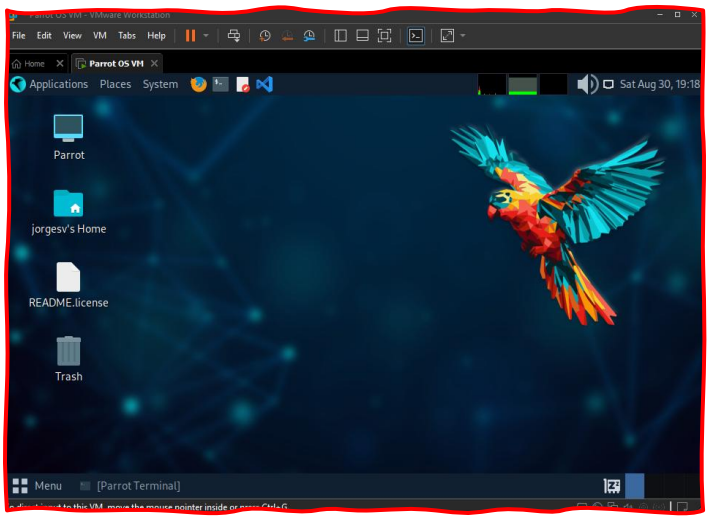
Step 2: Follow Installation Steps

1. Select the installation language, location, and keyboard layout.
2. Follow the on-screen instructions to partition the virtual disk (use the default settings for simplicity).
3. Set up a root password and create a non-root user account for daily use.

- **Root Password:** Password01
- **Non-Root User Account:** either your name or your atlas ID
- **Non-Root Password:** Password01

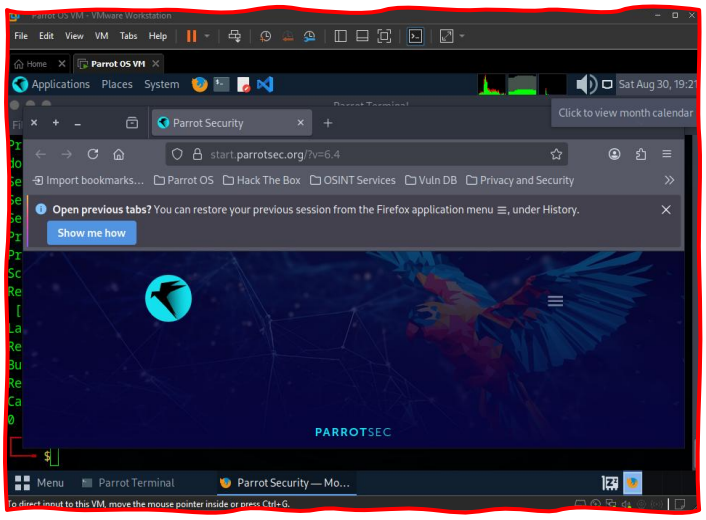
Step 3: Complete the Installation

1. Once installation is complete, reboot the VM.
2. Remove the ISO from the virtual CD/DVD drive to avoid booting into the installer again.
   - VM → Settings → CD/DVD (SATA) → Use Physical Drive
3. Log back into your machine. Take a screenshot of the desktop:



Part 4: Post-Installation Tasks

1. Log in using the non-root user account.
2. Update the system by running the following commands in the terminal**:
   ```
   sudo apt update && sudo apt-get update
   ```
3. Test network connectivity by opening a browser in Parrot OS and visiting a website (e.g., https://parrotsec.org).
4. Take a screenshot of the webpage open on your machine:



**If you have the "NO_PUBKEY 7A8286AF0E81EE4A" issue with the update running, do the following:

1. Open Firefox on your Parrot machine, and go to: https://deb.parrot.sh/parrot/pool/main/p/parrot-archive-keyring/parrot-archive-keyring_2024.12_all.deb
2. This will download a package with the updated key. To extract and install the key, run the following command:

```
sudo dpkg -i parrot-archive-keyring_2024.12_all.deb
```

3. Now, try running the update again.

Reflection Questions

1. Why is it important to verify the ISO checksum before installation?
   It is important to verify the ISO checksum to ensure its authentic and original. This ensures the file wasn't tampered with during any point during the download.
2. What are the benefits of using a virtual machine for cybersecurity labs?
   Using VMs ensures you do not do catastrophic damage to your pc OS. VMs also give you the ability to practice pentesting and other forms of testing. It is also isolated and can be accessed anywhere that has access to an internet source.
3. What type of distribution is Parrot OS?
   Debian based OS.